

How To: Software Engineer

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iOS Application
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I. Executive Summary

Becoming a software engineer can seem like an overwhelming feat for many. Being an employed software engineer at a tech company means that you have completed a rigorous curriculum regardless of educational background or institution. For others, being an employed software engineer meant completing a multi-month coding boot camp or enduring intense self-discipline necessary for becoming a self-taught programmer. In either case, a general career path is necessary in order to reach the goal of officially becoming a software engineer. “*How To: Software Engineer*”, an iOS application, aims to educate users by providing users with computer science / software engineering resources. In addition, registered users can supplement their path into software engineering with our mentor / mentee matching process. All users will find a wealth of positive reaffirming, informative, and organized resources to aid their journey to becoming a software engineer.

II. Use Cases

1. Guest User

Mia is an 8th grader in the Oakland Unified School District. Mia has just become interested in possibly becoming a software engineer later in life after a software engineer visited her computer science class. Due to her newfound interest, Mia downloads the “*How To: Software Engineer*” application from the iOS App store. Mia quickly browses the app to learn more about the career path and find supplemental computer science videos to aid her learning about the subject. After viewing the app, Mia becomes more informed about what it takes to become a software engineer and begins to craft her path to her new goal.

2. Registered User

Janie is an interior designer in Miami, Florida. Janie has been planning to make a career change into high technology. Janie's goal is becoming a software engineer in a matter of months. Janie is currently enrolled at a coding boot camp and is learning great amounts of information very quickly. In addition to learning the curriculum Janie desires to have a like-minded mentor in her future industry. Janie logs into the "*How To: Software Engineer*" app and begins browsing through potential mentors. Janie successfully matches with a mentor and begins conversation about her goals, future, and current challenges. Janie and her mentor build a relationship and is eventually offered a chance to interview at her mentor's company.

3. Admin User

Kim is the administrator of the "How To: Software Engineer" app. Kim's responsibility involves reviewing the mentor / mentee profiles and verifying that the profiles follow app guidelines and do not violate terms and conditions.

III. Data Definitions

1. Guest User

Any guest user visiting the application. Guest users can browse resources but cannot create a mentee profile or request a mentor.

2. Registered User

Any user who has created an account within the application. Registered users have all guest user privileges in addition to being able to create a mentee / mentor profile and request a mentor.

- Username: Name associated with person (First & Last Name given upon sign up)
- Email: Email address associated with user
- Password: Secret credentials that gives user access to their account

3. Admin User

A special user account. Admin user account can remove mentee / mentor profiles that violate app guidelines, terms and conditions.

4. Mentee / Mentor Profile

A profile consisting of:

- Profile photo
- First Name
- Last Name
- Occupation / Industry / School
- Location
- Brief background of user
- Skills (Can include soft or hard (technical) skills)
- Desired Skills

IV. Functional Requirements

1. Guest User

- Guest users shall be able to search all resources within resource view
- Guest users shall be able to view all resources within resource view

2. Registered User

- Registered users shall be able to search all resources within resource view
- Registered users shall be able to view all resources within resource view
- Registered users shall be able to create mentee / mentor profile
- Registered users shall be able to view other mentee / mentor profiles
- Registered users shall be able to message / interact with other mentees / mentors

3. Admin User

- Admin users shall be able to remove mentee / mentor profiles within violation of app guidelines, terms, and conditions.

V. Non-Functional Requirements

- Application shall be media rich (containing images, videos, articles, and books obtained from YouTube, Good Reads, and Unsplash API's)
- Privacy of users shall be protected and all privacy policies will be appropriately communicated to users
- The initial language used throughout the application shall be English
- Application security: Basic best practices shall be applied

VI. High-Level System

1. Design Pattern

- MVC Architecture Pattern

2. Front End

- Swift
- Foundation Framework
- Core Data Framework and API's
- Cocoa Touch Framework

3. Back End

- BaaS (Firebase)

4. Version Control

- GitHub / Git

Firebase NoSQL Schema (Tentative)

- Users (Mentor Example)
 - Unique ID (userid)
 - First Name: John
 - Last Name: Gosh
 - E-mail Address: JohnGosh@gmail.com
 - School / Occupation: Software Engineer
 - Location: San Francisco / Bay Area

- Skills: iOS, HTML5, CSS, Redux, React, Java, C++, Django,
- Desired Skills: Mentoring
- Users (Mentee Example)
 - Unique ID (userid)
 - First Name: Stacey
 - Last Name: Star
 - E-mail Address: StaceyStar@gmail.com
 - School / Occupation: Student @ San Francisco State University
 - Location: San Francisco / Bay Area
 - Skills: Python, C, C#, Go
 - Desired Skills: HTML5, CSS, Redux, React, Django

VII. Mock Ups



Logo Flash Screen



Landing/Login Screen

Registration Page

02:30 PM

Create an Account

First Name *

Last Name *

Email Address *

Password *

School / Occupation

Location

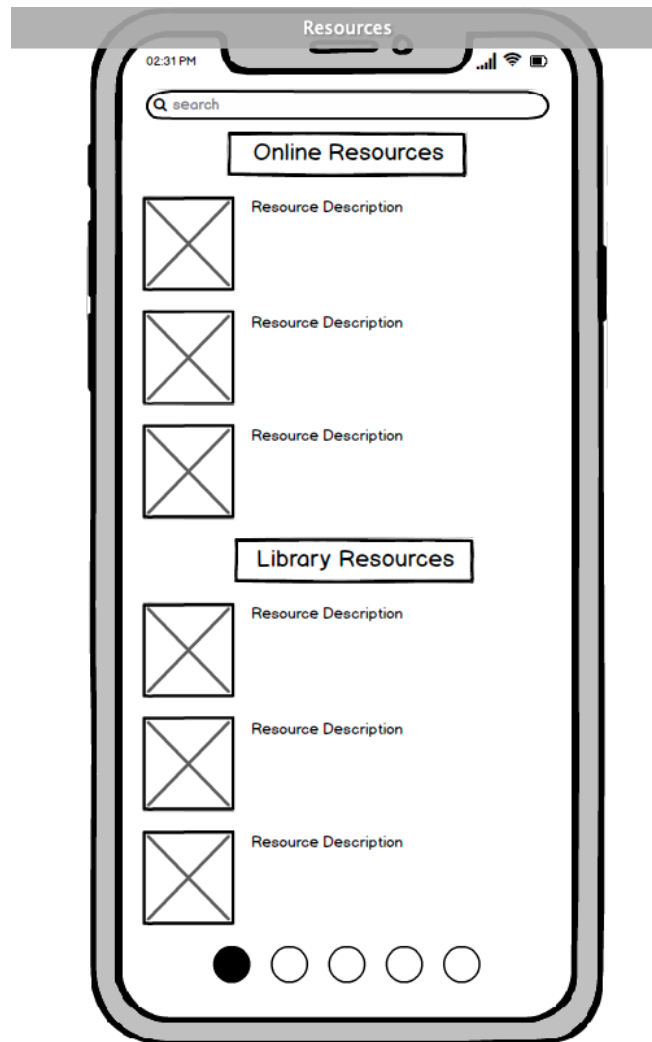
Skills

Desired Skills

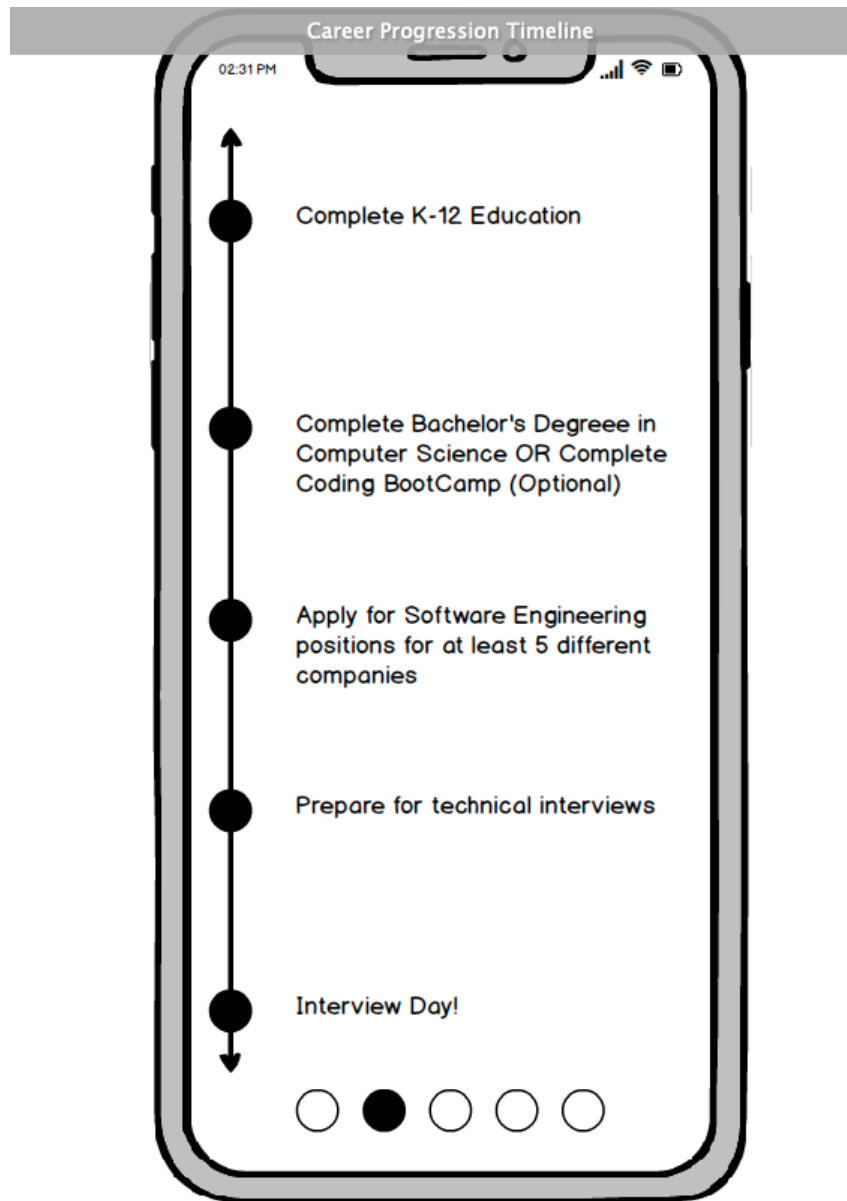
< Back

Sign Up

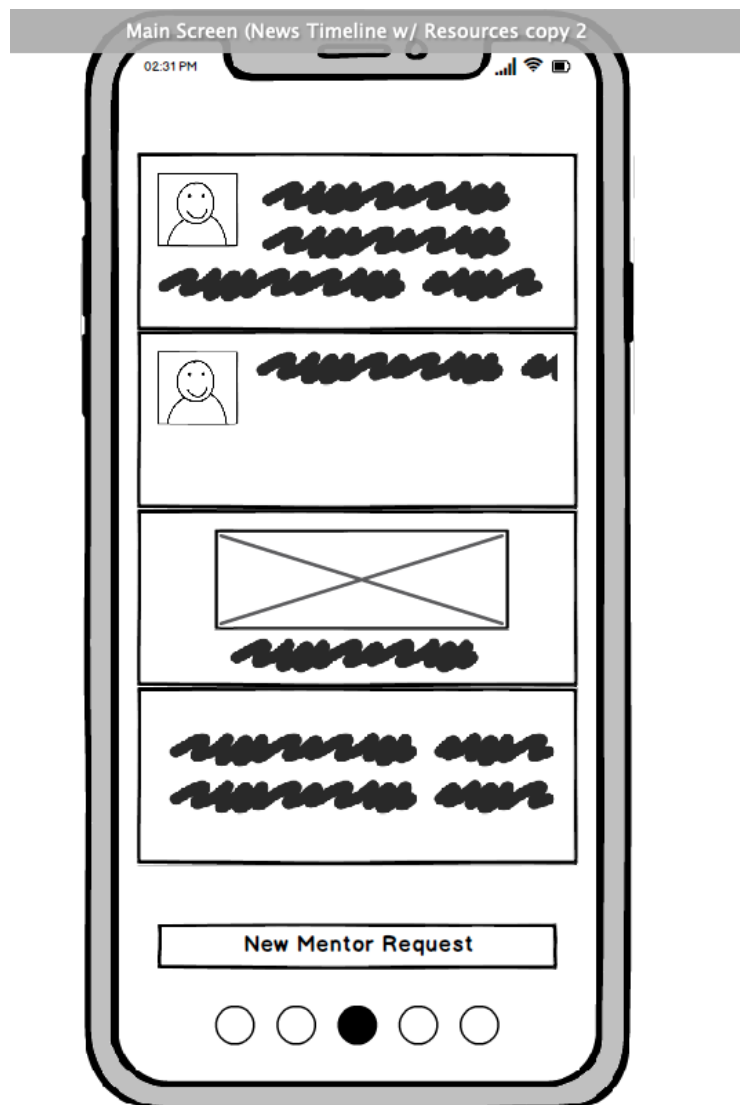
Registration Page



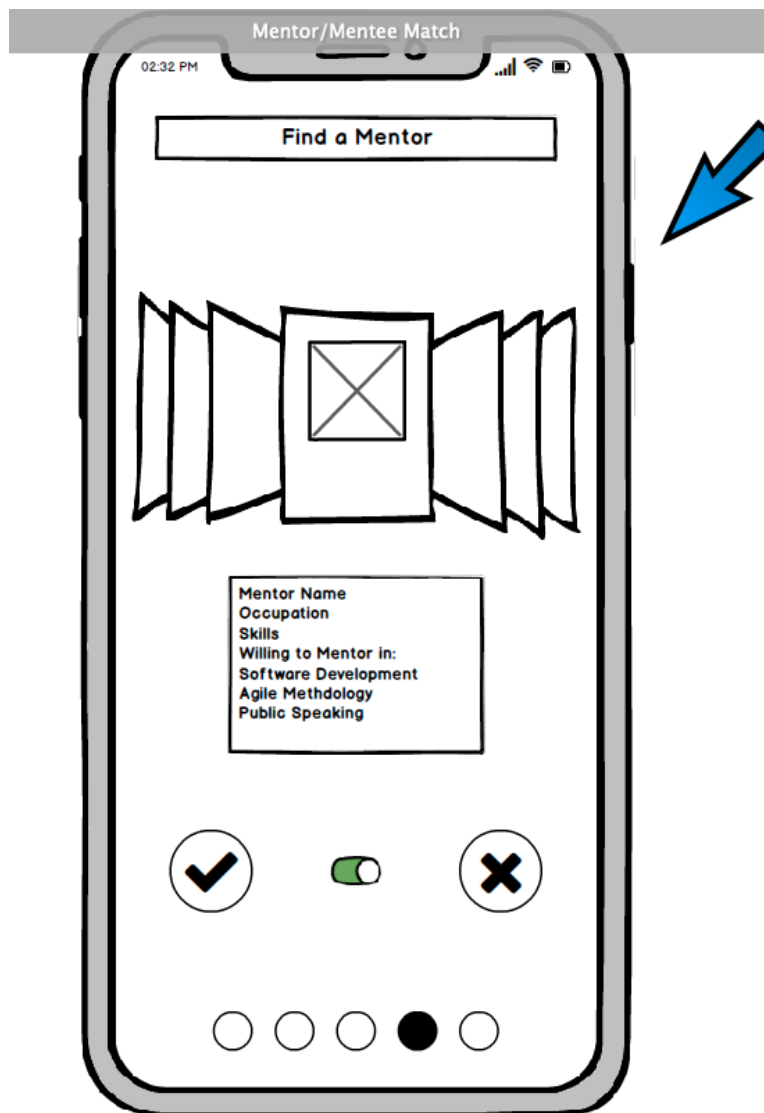
Resource Page



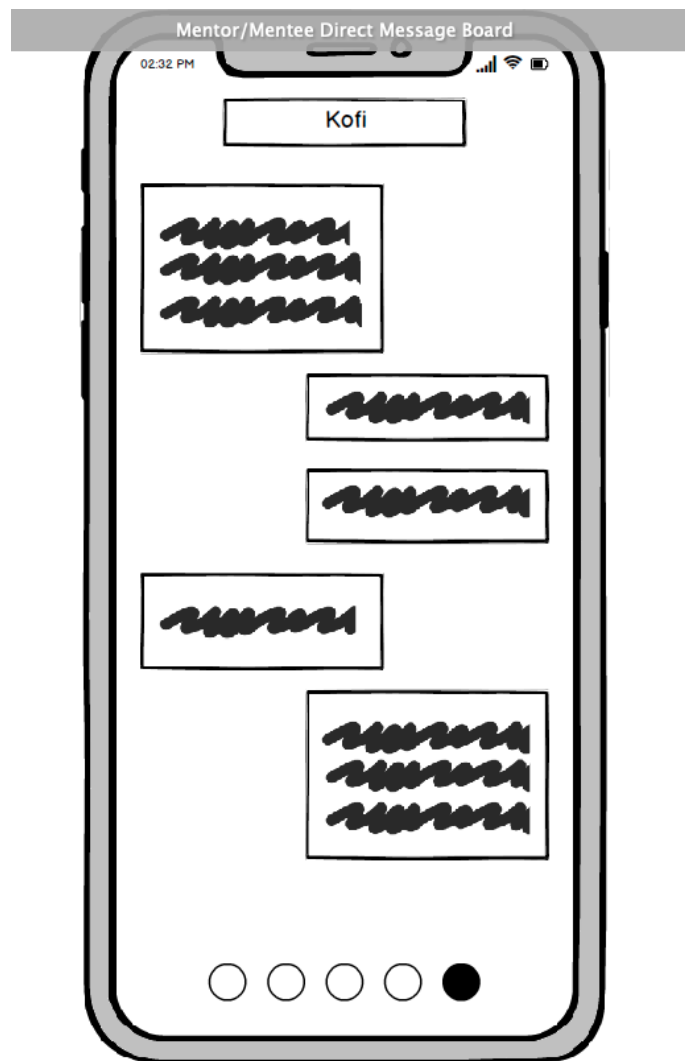
Career Progression Timeline Page



Main Screen (News Timeline w/ Resources) Page



Mentor/Mentee Match Page



Mentor/Mentee Direct Message Board